## MC 451 Describing Data in R rubric

**75 points.** Due Friday, November 20, 11:59 PM.

**How this is scored.** Add the points from each row; the total is out of 75. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Length: 300 to 600 words of your own writing (interpretation and alt text), not counting code or output. Word counts cover your own writing, not the title, headings, tables, references, figure captions, code, or output, and writing more than the range never costs points. Scores on work that shows effort are adjusted with the course's curve: your final percentage is 25 percent plus 75 percent of your rubric percentage, so 50 of 100 on the rubric becomes 62.5. Work at two-thirds or more of the minimum length counts as showing effort. Submissions are unlimited before the deadline, so you can revise and submit again.

- The worksheet is 12-describing-data-mc451.qmd. It loads `twitch_analysis.RDS` with your hand-coded variable inside, writes the code that is not the point, and leaves blanks for the tables and figures and a Write prompt after each output. The Describing Data risks guide shows the traps on this job and how to check for each.
- The calls these tables and figures lean on, all of which run the same in VS Code: `count()` for a frequency table, `pivot_wider()` to lay a cross-tabulation out, `fct_reorder()` to order the bars, `labs()` for the axis labels, title, and caption, `scale_fill_brewer()` or the v2v palette for the fill, and position = "fill" with `scales::percent_format()` for a proportion chart. The Describe group of the R Companion's Write R code writes these same calls, including Line chart over time and Histogram comparing two groups, saves each figure with `ggsave()`, lays the crosstab out with `count()` and `pivot_wider()`, leaves an alt-text line marked YOUR TURN in each chart for you to write, and opens the saved chart in a VS Code tab.
- What it feeds: your White Paper, which runs these same table and figure calls on your own variables. The Twitch figures stay here; the paper shows only what your own data shows.

*Describing Data in R, 75 points*

| Criterion | Exemplary | Proficient | Developing | Beginning |
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| **Frequency tables**<br>15 points | **15 points.** A frequency table for each key categorical variable, your hand-coded variable among them on its thirty messages, built with `count()` and labeled so a reader can tell which variable each one shows. | **11 to 14 points.** Tables for most key variables, or a table without labels. | **8 to 10 points.** One table, or tables printed as raw R output with no labels. | **0 to 7 points.** None. |
| **The three numbers**<br>15 points | **15 points.** Mean, median, and standard deviation for every continuous variable you report, and a sentence whenever the mean and median disagree. | **11 to 14 points.** All three numbers reported, but a large gap between the mean and the median goes unremarked. When they part company, the distribution is telling you something. | **8 to 10 points.** The mean alone, or the three numbers for only some variables. | **0 to 7 points.** No summary numbers. |
| **Figures**<br>20 points | **20 points.** At least two ggplot2 figures plus one figure of your own choice, each with axis labels, a title, and a caption, saved to a file by the code that makes them, with one sentence per figure disclosing a choice you made (a bin width, a cap on an axis, a category you merged), and one sentence on why you chose the third figure for that variable. | **15 to 19 points.** The figures are there, but one is missing a label, the title, the caption, or its disclosure sentence, or the figure you chose has no reason given. | **10 to 14 points.** One figure, or figures saved by hand instead of by the code. | **0 to 9 points.** No figures. |

| Criterion | Exemplary | Proficient | Developing | Beginning |
|---|---|---|---|---|
| **Cross-tabulation**<br>10 points | **10 points.** A cross-tabulation of two categorical variables, laid out with `count()` and `pivot_wider()`, labeled so a reader can tell the rows from the columns. This is the table next week's chi-square tests. | **8 to 9 points.** Present, but the labels are missing. | **5 to 7 points.** Built on the wrong columns (a variable crossed with itself, or an ID column). | **0 to 4 points.** Missing. |
| **Alt text**<br>5 points | **5 points.** Every figure has alt text that says what the figure shows, not a repeat of its caption. | **4 points.** Alt text on every figure, but one repeats the caption. | **3 points.** Alt text on some figures only. | **0 to 2 points.** None. |
| **Interpretation**<br>10 points | **10 points.** Two to three sentences at each Write prompt, saying what the output shows and what it does not. | **8 to 9 points.** Sentences at each prompt, but they describe the output without saying what it cannot show. | **5 to 7 points.** Interpretation at some prompts only, or a single line each, which usually means the "what it does not show" half is missing. | **0 to 4 points.** None. |